

Metadata-Guided Cross-Attention Fusion for Multimodal Skin Lesion Classification: A Leakage-Audited, Cross-Dataset Validated Approach

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Abstract—Skin lesion classification systems often report inflated performance due to data leakage, image-wise (non-patient-wise) splitting, and evaluation on a single dataset. We present a multimodal fusion architecture that combines dermoscopic/clinical images with structured patient metadata via cross-attention, where metadata queries guide the model’s attention over spatial image features. Our pipeline includes a systematic leakage audit (22 columns excluded across 4 datasets, each backed by statistical evidence), patient-wise splitting, and evaluation on a held-out test set touched exactly once. On PAD-UFES-20 (6-class, macro-F1), our cross-attention fusion model achieves 0.6977 test macro-F1, significantly outperforming late fusion and single-modality baselines (all significant, Bonferroni-corrected). Cross-dataset evaluation on HAM10000 and ISIC archives, bootstrap significance testing, and Fitzpatrick skin-tone fairness analysis further validate the model’s generalization and equity properties. We additionally report extended ablations (backbone comparison, dataset expansion, ensemble, joint three-way fusion) under a pre-registered decision protocol, providing a transparent, reproducible account of what improved performance and what did not. Our results demonstrate that methodological rigor and honest reporting are as important as raw accuracy for clinically deployable dermatology AI.

Index Terms—multimodal fusion, cross-attention, skin lesion classification, data leakage, cross-dataset generalization, fairness

I. INTRODUCTION

Skin cancer and other dermatological conditions are among the most prevalent diseases worldwide, and early, accurate diagnosis substantially improves patient outcomes, particularly since visual inspection remains the first and most accessible point of clinical contact in many healthcare settings. Automated screening tools deployed alongside, rather than in place of, dermatologist judgment therefore have direct practical value, especially where specialist access is limited. Computer-aided diagnosis (CAD) systems built on deep learning achieve strong performance on dermoscopic image classification; however, most rely on image features alone, despite dermatologists routinely incorporating patient history – age, lesion duration, itching, prior growth, anatomical site – into their reasoning. Metadata of this kind is frequently available in public dermatology datasets but remains underused as a modeling signal.

A second, equally important limitation concerns evaluation rigor. Many reported results are obtained under conditions that inflate apparent performance: binary framing (malignant versus benign) rather than the clinically relevant multi-class problem; accuracy as the primary metric, which can mask poor

performance on minority classes in imbalanced data; absence of a documented leakage audit; and evaluation confined to a single dataset, which does not test whether a model generalizes beyond the imaging conditions, patient population, and label-collection procedure of its training source.

A third, frequently overlooked limitation concerns class imbalance. Public dermatology datasets are dominated by common, low-risk classes, while clinically urgent classes such as Melanoma are severely underrepresented. Many prior works address this purely through synthetic augmentation, which adds no genuinely new diagnostic signal. We instead source additional real, biopsy-confirmed images for the weakest classes from independent public collections, and empirically isolate how much this data-level intervention contributes relative to architectural improvements.

Beyond the modeling question, a growing body of work raises concerns about equity in dermatology AI: models trained predominantly on lighter skin tones can silently underperform on darker skin tones, a risk rarely measured, let alone reported, in typical benchmark papers. A trustworthy skin lesion classifier therefore needs to be evaluated not just for accuracy, but for the integrity of its evaluation protocol and its behavior across sub-populations.

This work addresses these limitations directly, proposing a metadata-guided cross-attention fusion architecture paired with a rigorous, transparently documented evaluation protocol. Our contributions are as follows:

- 1) A cross-attention fusion architecture in which patient metadata queries spatial image tokens, avoiding the dimensional imbalance that undermines naive late-fusion approaches.
- 2) A systematic, statistically-grounded leakage-audit methodology, applied across four datasets, identifying and excluding twenty-two metadata columns that would otherwise allow the model to exploit post-diagnosis information.
- 3) A targeted, real-data class-imbalance correction for the two rarest diagnostic classes, isolated experimentally from architectural effects.
- 4) Cross-dataset generalization testing (PAD-UFES-20 to HAM10000) and external validation on two ISIC archives, quantifying how performance transfers beyond the training distribution.

- 5) Bootstrap-based statistical significance testing for cross-dataset and extended-ablation comparisons, and a Fitzpatrick skin-tone fairness analysis.
- 6) A pre-registered protocol for extended architecture and data ablations, reported honestly, including results that did not exceed the primary model.

The remainder of this paper is organized as follows. Section II reviews prior machine learning and deep learning approaches to skin lesion classification. Section III describes our datasets, leakage-audit methodology, and proposed architecture. Section IV presents our experimental results, including cross-dataset generalization, fairness analysis, and extended ablations. Section V discusses the implications and limitations of our findings, and Section VI concludes with directions for future work.

II. RELATED WORK

A. Machine Learning-Based Approaches

Early computer-aided diagnosis systems for skin lesions relied on handcrafted feature extraction (color histograms, texture descriptors, border irregularity measures) combined with classical classifiers. More recent tabular-only approaches use structured clinical data directly instead of hand-engineered image features. The TRACE framework applies transformer-based modeling to structured patient metadata, showing clinical fields alone carry substantial diagnostic signal without image input. However, TRACE’s PAD-UFES-20 evaluation uses a 90/10 train-validation split with no held-out test set, so its reported DANET (0.625) and TRACE (0.783) macro-F1 figures are validation-style estimates, not results comparable to a true test-set score.

B. Deep Learning-Based Approaches

The dominant paradigm fine-tunes CNNs (e.g., ResNet, EfficientNet) on dermoscopic images, often without leveraging clinical metadata. More recently, vision-language approaches have entered the field: Shrestha and Palit (2026) combine MedCLIP-derived image-text embeddings with patient metadata through early and attention-based fusion on a curated ISIC 2024 subset, reporting 96% accuracy and an AUROC of 0.987. This is a binary-classification, accuracy-based result and, like the tabular comparators above, not directly comparable to our multi-class macro-F1 evaluation, though it illustrates the value of combining vision-language representations with metadata.

C. Multimodal Fusion Mechanisms

A smaller body of work directly addresses multimodal image-metadata fusion mechanisms. Pacheco and Krohling’s MetaBlock, from PAD-UFES-20’s creators, modulates image feature channels via a metadata-conditioned gating function applied uniformly across spatial locations – unlike this work’s per-location spatial cross-attention. The same authors’ earlier clinical-impact study found Squamous Cell Carcinoma and Basal Cell Carcinoma share near-identical clinical profiles and remain confused even with metadata, corroborated by our

TABLE I
COMPARISON WITH RELATED WORK

Aspect	Our Work	TRACE	Meta-Block	Shrestha & Palit
Approach type	DL, multi-modal	ML, tabular	DL, multi-modal	DL, VLM+meta
Modality	Image+Meta	Metadata only	Image+Meta	Image+Meta (VLM)
Classes	6	N/D ^a	N/C ^b	2 (binary)
Metric	Macro-F1	Macro-F1	Acc. ^c	Accuracy
Held-out test	Yes	No (val-only)	N/C ^b	Yes
Leakage audit	22 cols.	Not reported	N/C ^b	Not reported
Cross-dataset	Yes	No	N/C ^b	No
Fairness	Yes	No	N/C ^b	No

^aN/D: not documented. ^bN/C: unconfirmed (paywalled). ^cSecondary-source, unverified.

own confusion analysis (Section IV-A). Comparative studies of fusion strategies on HAM10000 independently found cross-attention (image=K/V, metadata=Q) outperforms naive concatenation and self-attention, corroborating our architectural choice. More recently, Christopoulos et al.’s SLIMP applies self-supervised image-metadata contrastive pretraining on PAD-UFES-20, reaching 0.825 macro-F1 – a realistic upper bound achieved via a more resource-intensive pipeline than our supervised baseline. Yan et al.’s PanDerm, a foundation model pretrained on two million clinical images, achieves strong few-label results but is not yet evaluated under a comparable leakage-audited protocol.

D. Fairness in Dermatology AI

Fairness in dermatology AI has received growing attention following Daneshjou et al.’s (2022) demonstration that dermatology AI models can suffer a 27–36 percentage point drop in ROC-AUC when evaluated on the Diverse Dermatology Images dataset, particularly for darker Fitzpatrick skin types and uncommon diseases. Alipour et al. (2024) subsequently showed that this is a systemic reporting and representation gap across public dermatology datasets rather than an isolated finding. Xu et al. (2024) found reduced per-Fitzpatrick sensitivity even at the pretrained-embedding level of foundation models – a caution for any future foundation-model extension of this work. To our knowledge, no prior work combines a statistically-audited leakage methodology, cross-dataset generalization testing, bootstrap significance testing, and Fitzpatrick fairness analysis within a single, transparently reported pipeline. Table I summarizes these comparisons.

III. METHODOLOGY

Figure 1 presents an overview of our dataset preparation pipeline, from raw data sourcing through the leakage-audited, patient-wise-split processed output, before we describe each phase in detail.

A. Datasets

Table II summarizes the four datasets used in this study. PAD-UFES-20 (2,298 images, six diagnostic classes) serves as the primary dataset for training and final evaluation, and is the only dataset among those used that provides rich structured

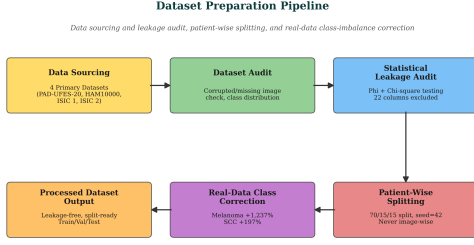


Fig. 1. Dataset preparation pipeline: data sourcing, quality audit, statistical leakage audit, patient-wise splitting, real-data class-imbalance correction, and the resulting processed dataset.

TABLE II
DATASET OVERVIEW

Dataset	Images	Classes	Role
PAD-UFES-20	2,298	6	Primary train/val/test
HAM10000	10,015	7	Cross-dataset generalization
ISIC Archive 1	2,357	9	External validation
ISIC Archive 2	25,076	9	External validation

clinical metadata (twenty-one usable features after audit) – a key distinguishing property of our data foundation, since it is what makes genuine image+metadata fusion possible at all. HAM10000 (10,015 images, seven classes) is used exclusively as a cross-dataset generalization target. ISIC Archive 1 (2,357 images) and ISIC Archive 2 (25,076 images) are used solely for external validation and were never included in any training set.

To address severe class imbalance in the two rarest PAD-UFES-20 categories, we incorporated biopsy-confirmed images from two external, non-overlapping public sources, expanding Melanoma from 38 to 508 images (a 1,237% increase) and Squamous Cell Carcinoma from 135 to 401 images (a 197% increase), as shown in Figure 2. Unlike synthetic augmentation, every added image is a real, independently-sourced, biopsy-confirmed case, verified to have zero patient overlap with any evaluation split. This expanded variant is used only in the later ablation experiments (Section IV-E), never in the primary reported result, preserving the integrity of our locked headline evaluation.

B. Data Leakage Audit

Prior to model development, every metadata column in each dataset was tested for statistical association with the diagnostic label using the phi coefficient and chi-square test. Columns exhibiting near-deterministic association with the label – indicating that the field encodes post-diagnosis information rather than pre-diagnosis clinical observation – were excluded. The most severe example was PAD-UFES-20’s biopsied field, which was True in 100% of malignant cases ($\phi = 0.80$), reflecting the clinical practice of biopsying only lesions already suspected malignant. This audit identified and excluded twenty-two columns across the four datasets, yielding a documented feature whitelist per dataset (twenty-one features for PAD-UFES-20).

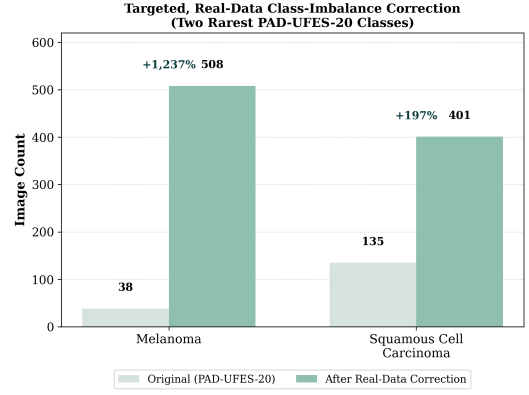


Fig. 2. Targeted, real-data class-imbalance correction for the two rarest PAD-UFES-20 classes, used only in later ablation experiments.

C. Data Splitting

All splits were performed at the patient level, never at the image level, to prevent images from the same individual appearing in both training and evaluation partitions. PAD-UFES-20 was split 70/15/15 with a fixed seed (42) for reproducibility. The test partition was evaluated under a single-use guard mechanism that raises an error on any repeated access.

D. Proposed Hierarchical Architecture

Figure 3 details our proposed hierarchical architecture as a nine-stage pipeline. The image branch is an EfficientNet-B0 convolutional network, initialized with ImageNet-pretrained weights and fully fine-tuned; its pre-pooling feature map yields forty-nine spatial tokens. One-hot encoding the 21 whitelisted PAD-UFES-20 clinical fields (Section III-B) yields an 89-dimensional input vector, which the metadata branch – a multilayer perceptron – maps to a 64-dimensional embedding. Before cross-attention, a metadata-conditioned channel gate (MetadataChannelGate) modulates the 49 image tokens: a sigmoid function of the metadata embedding produces a 1280-dimensional gate applied uniformly across all spatial tokens’ channels, allowing clinical context to influence low-level image features prior to the attention-based fusion step. Both raw representations are then projected into a shared 256-dimensional space: the gated image tokens through a Key/Value projection, metadata through a Query projection. An 8-head cross-attention module then fuses them – the metadata Query attends over the image Key/Value tokens, producing a 256-dimensional attended vector. This is concatenated with the original 64-dimensional metadata embedding into a 320-dimensional joint representation, which is passed through a classification head ($\text{FC}(128) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{FC}(6)$) to produce the final six-class prediction. Both branches are warm-started from independently pretrained single-modality checkpoints and subsequently fine-tuned end-to-end.

We selected cross-attention over simple feature concatenation (late fusion) because the latter suffers from a dimensional imbalance: a 1,280-dimensional image embedding numerically dominates a 64-dimensional metadata embedding when directly concatenated, causing the fused representation to be

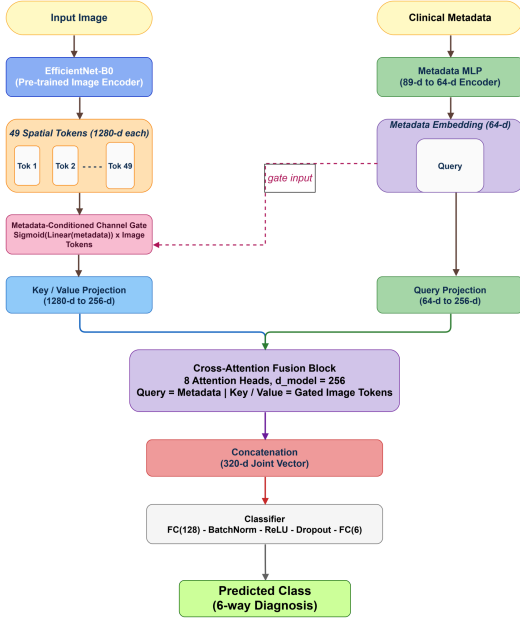


Fig. 3. Proposed hierarchical cross-attention fusion architecture. Image tokens pass through a metadata-conditioned channel gate before image and metadata paths are projected into a shared 256-dimensional space, fused via an 8-head cross-attention block ($Q=\text{metadata}$, $K=V=\text{gated image tokens}$), and classified through a joint head.

driven almost entirely by the image modality. Projecting both modalities into a shared attention space before fusion avoids this imbalance, and allows the model to learn which spatial regions of the lesion are most relevant given a specific patient’s clinical profile.

E. Training Configuration

All models were trained with the Adam optimizer (learning rate 1×10^{-5} for fusion/cross-attention models, 1×10^{-4} for image-only, 1×10^{-3} for metadata-only; batch size 32; weight decay 1×10^{-4} ; up to 30 epochs), class-weighted cross-entropy loss, three random seeds per configuration, and early stopping (patience of seven epochs) based on validation macro-F1. Images were resized to 224×224 (aspect-preserving, zero-padded); no augmentation was applied. Missing categorical metadata was encoded as an explicit “missing” category (not imputed). Training used a single NVIDIA T4 GPU (Kaggle).

IV. EXPERIMENTS AND RESULTS

A. Primary Results

Table III reports validation and test macro-F1 for the four primary model variants on PAD-UFES-20. The cross-attention fusion model achieves the best performance on both partitions, with a test macro-F1 of 0.6977, a 9-percentage-point improvement over the metadata-only baseline. The same test predictions also give a mean accuracy of 0.763; we report macro-F1 as the headline metric throughout, since it weights all six classes equally, whereas accuracy is dominated by the majority classes.

Table IV shows the test-set confusion matrix (summed over 3 seeds, 1,062 predictions). The dominant error cluster is among Actinic Keratosis (AK), Basal Cell Carcinoma (BCC),

TABLE III
PRIMARY RESULTS ON PAD-UFES-20 (MACRO-F1, MEAN OVER 3 SEEDS)

Model	Validation	Test
Image-only	0.5703 ± 0.0130	0.6175 ± 0.0153
Metadata-only	0.5762 ± 0.0072	0.6077 ± 0.0202
Late Fusion	0.5731 ± 0.0021	0.6566 ± 0.0234
Cross-Attention	0.6209 ± 0.0143	0.6977 ± 0.0269

Paired bootstrap (1,000 resamples), 95% CI: cross-attention’s gain is significant vs. every baseline – image $p < 0.001$ [0.051, 0.117], metadata $p = 0.006$ [0.031, 0.151], late fusion $p = 0.002$ [0.018, 0.067] (all sig. under Bonferroni $\alpha = 0.0167$). Reported as mean per-seed macro-F1; Table V’s macro-average (0.6975) instead uses the pooled 3-seed confusion matrix – related but distinct, not a discrepancy.

TABLE IV
CONFUSION MATRIX, CROSS-ATTENTION MODEL, PAD-UFES-20 TEST (SUMMED, 3 SEEDS)

	AK	BCC	MEL	NEV	SEK	SCC
AK	268	22	0	0	17	20
BCC	30	328	0	4	3	34
MEL	0	0	16	8	0	0
NEV	1	8	1	87	8	0
SEK	3	4	2	12	89	1
SCC	15	51	1	0	7	22

and Squamous Cell Carcinoma (SCC) – three keratinocyte-lineage, actinic-damage-spectrum classes that account for 172 of 252 total misclassifications (68.3%), with $\text{SCC} \rightarrow \text{BCC}$ (51 cases) the single largest confusion. This corroborates Pacheco and Krohling’s (2020) clinical finding that these classes remain difficult to separate even with metadata available. Per-class performance mirrors this pattern: Actinic Keratosis reaches $F1 = 0.83$ (precision 0.85, recall 0.82) and Basal Cell Carcinoma $F1 = 0.81$ (precision 0.79, recall 0.82), while Melanoma reaches $F1 = 0.73$ (precision 0.80, recall 0.67) despite only 24 test instances, and Squamous Cell Carcinoma – the class most entangled with BCC – is the weakest at $F1 = 0.25$ (precision 0.29, recall 0.23), consistent with its severe test-set scarcity (32 images) and motivating the targeted class-imbalance correction in Section III-A.

B. Cross-Dataset Generalization

We evaluated PAD-UFES-20-trained models on HAM10000’s three shared diagnostic classes (Basal Cell Carcinoma, Melanoma, Nevus). Table VI reports the resulting macro-F1 scores. Cross-attention fusion (0.4654) is not statistically distinguishable from image-only (0.4658, $p = 0.970$, 95% CI $[-0.019, 0.020]$) or late fusion (0.4597, $p = 0.590$, CI $[-0.014, 0.025]$), but significantly outperforms metadata-only (0.2920, $p < 0.001$, CI $[0.134, 0.210]$) – reflecting HAM10000’s sparser metadata schema. We report this honestly: cross-attention demonstrably wins the modality-value question but does not yet demonstrate superior pure generalization relative to image-only features.

C. External Validation and Fairness Analysis

HAM10000-trained checkpoints were further evaluated on ISIC Archive 1 and Archive 2 without additional fine-tuning. Because HAM10000 shares substantial image provenance with both archives (98.6% overlap with Archive 2, 66.5% with Archive 1), overlapping images were identified and excluded from the evaluation sets via a dedicated exclusion list (1,362 IDs), so the reported numbers reflect genuinely unseen data.

TABLE V
PER-CLASS PRECISION, RECALL, AND F1 (CROSS-ATTENTION MODEL,
SUMMED 3 SEEDS)

Class	Support	Precision	Recall	F1
AK	327	0.8454	0.8196	0.8323
BCC	399	0.7942	0.8221	0.8079
MEL	24	0.8000	0.6667	0.7273
NEV	105	0.7838	0.8286	0.8056
SEK	111	0.7177	0.8018	0.7574
SCC	96	0.2857	0.2292	0.2543
Macro avg	1062	0.7045	0.6947	0.6975

TABLE VI
PAD-UFES-20 $\rightarrow$ HAM10000 CROSS-DATASET MACRO-F1

Model	Macro-F1
Image-only	0.4658 $\pm$ 0.0373
Metadata-only	0.2920 $\pm$ 0.0121
Late Fusion	0.4597 $\pm$ 0.0084
Cross-Attention Fusion	0.4654 $\pm$ 0.0197

Image-only macro-F1 was 0.4912 ± 0.0094 on Archive 2 and 0.2421 ± 0.0118 on Archive 1 (the latter driven by near-zero support for two of five shared classes, not worse generalization). On Archive 2, image-only significantly outperformed metadata-only (0.2410 ± 0.0295 ; bootstrap diff $+0.2502$, 95% CI $[+0.2368, +0.2641]$, $p < 0.001$), the largest, most statistically decisive effect in this study. A Fitzpatrick skin-tone fairness analysis on PAD-UFES-20’s test partition found cross-attention fusion to be the best- or tied-best-performing model within every Fitzpatrick group with sufficient sample size for evaluation; however, the two darkest categories (Types V and VI) contained only two and one test images respectively, and 38% of test records lacked a recorded Fitzpatrick value. We report this as a genuine data-availability limitation rather than a definitive fairness conclusion, consistent with Alipour et al.’s (2024) finding that skin-tone underrepresentation is systemic across public dermatology datasets.

D. Extended Ablations

Motivated by the observation that our reported macro-F1 is comparatively modest relative to some binary or accuracy-based literature results, we conducted four further, pre-registered ablations to determine whether performance could be meaningfully improved through legitimate architectural or data changes. Table VII summarizes these results.

The dataset-expansion-only ablation isolates the effect of adding real, biopsy-confirmed Melanoma and Squamous Cell Carcinoma images (Figure 2) without changing the model architecture; its near-identical result to the original baseline indicates that the larger performance gains observed with backbone substitution are attributable primarily to architectural capacity and representation quality, not to training-set size alone – despite the 1,237% and 197% class-size increases shown above. For the dual-backbone ensemble, we had pre-registered a decision rule prior to observing results: a third, final read of the held-out test set would only be authorized if a candidate model’s validation score clearly and meaningfully exceeded the strongest single-backbone validation result (0.6710). The ensemble’s validation mean (0.6856) satisfied this criterion and was granted a one-time test evaluation, yielding 0.7321;

TABLE VII
EXTENDED ABLATION RESULTS (PAD-UFES-20)

Experiment	Val. macro-F1	Outcome
5-backbone comparison	0.5859–0.6224	Exceeded all single-modality baselines (max 0.5762)
Dataset-expansion only	0.6186 ± 0.0124	$\approx$ primary cross-attention result (0.6209)
Dual-backbone ensemble	0.6856; 0.7321(test)	$p = 0.062$, not sig.
Joint 3-way fusion	0.6721	Exceeded 0.6710 bar by only $+0.0011$, below clear-and-meaningful threshold
Ensemble+TTA	$0.6213 \rightarrow 0.5886^d$	Rejected: Melanoma F1 $0.3636 \rightarrow 0.20$

^dValidation macro-F1, before $\rightarrow$ after TTA.

bootstrap testing against the original headline result, however, returned $p = 0.062$ (95% CI $[-0.002, 0.077]$, straddling zero), narrowly missing the conventional 0.05 threshold. The joint three-way fusion model’s validation mean (0.6721) did not clearly exceed the same 0.6710 threshold and was therefore not granted a test evaluation, consistent with the pre-registered protocol. We report both outcomes transparently rather than selectively.

V. DISCUSSION

A. Interpretation of Results

Several methodological choices explain why our reported scores are lower than some literature values. We address a more difficult six-class problem (not binary) and report macro-F1, not accuracy, which weights every class equally rather than masking minority-class errors. Our leakage audit also removed twenty-two shortcut-prone features. Restricted to comparably rigorous multi-class, macro-F1 evaluation, our result compares favorably. The confusion matrix (Table IV) shows a genuine, clinically-explainable difficulty rather than an architectural weakness: the confused classes share the same actinic-damage-spectrum lineage that clinicians also find hard to distinguish. Unlike the cross-dataset comparisons (Section IV-C, all non-significant, $p \geq 0.59$) and the ensemble ($p = 0.062$), the primary in-distribution comparison is the one result that is both numerically consistent and statistically robust (all $p \leq 0.006$) – real within-distribution, even though it does not yet translate into superior cross-dataset generalization.

The dataset-expansion-only ablation is itself a notable finding: adding real training data for the weakest classes, without an architectural change, produced negligible gains, whereas substituting the image backbone alone produced substantial gains – suggesting model capacity, not raw training-set size, was the more binding constraint at this dataset scale.

The ensemble result (0.7321 test, $p = 0.062$) illustrates a further point: we observed consistent validation-to-test increases (primary model: $+0.077$; ensemble: $+0.047$), under which the joint fusion model’s test score might plausibly also have exceeded validation. We deliberately did not pursue this evaluation, since a post-hoc “it might do better” expectation would undermine pre-registering a threshold before observing results – a disciplined limitation, not an oversight.

We similarly rejected an ensemble-plus-TTA variant (Table VII, row 5): its per-class Melanoma F1 collapse, not the aggregate drop alone, is what we treat as clinically disqualifying.

B. Limitations

This study has several limitations that should inform how its results are interpreted. First, our fairness analysis is constrained by severe sample-size imbalance across Fitzpatrick skin-tone categories in PAD-UFES-20, particularly Types V and VI; findings should be read as suggestive rather than statistically conclusive for the darkest skin tones. Second, our cross-dataset evaluation is limited to three diagnostic classes shared between PAD-UFES-20 and HAM10000, since the two datasets do not share an identical taxonomy. Third, the dual-backbone ensemble and joint three-way fusion results, while promising, remain unconfirmed on a genuinely independent test set under our pre-registered protocol. Finally, all datasets originate from a small number of geographic and clinical contexts, and further external validation across more diverse populations and imaging equipment would strengthen real-world generalizability claims. We also did not profile inference latency or memory footprint (the two-backbone variants roughly double image-branch computation); a full deployment-readiness assessment is left to future work.

VI. CONCLUSION AND FUTURE WORK

We presented a metadata-guided cross-attention fusion architecture for multimodal skin lesion classification, achieving 0.6977 test macro-F1 on PAD-UFES-20 under a six-class, leakage-audited, patient-wise-split protocol. Beyond the primary result, we contribute a transparent, pre-registered methodology for extended architectural and data ablations, cross-dataset generalization testing, statistical significance testing, and skin-tone fairness analysis – demonstrating that methodological rigor and honest reporting of both positive and inconclusive findings are essential complements to raw predictive accuracy in clinically-oriented dermatology AI. Multimodal fusion substantially improves controlled, in-distribution performance, but rigorous cross-dataset evaluation reveals this does not automatically translate into domain robustness – underscoring that evaluation methodology matters as much as architecture.

Several directions follow directly from our findings. First, a fresh, independently-held-out evaluation set would allow the dual-backbone ensemble and joint three-way fusion results to be legitimately re-tested without compromising our current pre-registered protocol. Second, since the confusion matrix identifies Actinic Keratosis, Basal Cell Carcinoma, and Squamous Cell Carcinoma as a persistent, clinically-explainable error cluster, targeted data collection or a dedicated sub-classifier for this keratinocyte-lineage group is promising. Third, a multi-institution collaboration to collect labeled images across the full Fitzpatrick spectrum would enable a more statistically powered fairness evaluation, potentially using equalized-odds metrics beyond the per-group macro-F1 reported here. Finally, lesion segmentation as a pre-processing step, deferred here to

avoid confounding it with the ablations already in progress, remains a reasoned, sequenced direction for isolating any further gains it may offer.

ETHICS AND REPRODUCIBILITY STATEMENT

All datasets are publicly available and were de-identified by their publishers prior to release; the authors performed no patient contact, re-identification, or other IRB-requiring activity. Code, trained checkpoints, per-dataset feature whitelists, exact parameter counts, and per-seed training times will be publicly released upon acceptance; further hyperparameter details are available from the authors upon request.

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